MCO-1D

Flight Software Reference

mco_wave_library.py — complete system description

v25 · September 2026

Lines	**90,232**
Hypotheses	**162**
Genome axes	**15**
GS commands	**368**
ALC modalities	**3**
Autopilot deferred	**12**

1. Architecture

mco_wave_library.py is the single authoritative source for all flight kinematics, learning, and adaptive control. It runs on the flight computer at 50 Hz. All five learning subsystems (AI observer, discovery selector, ALSS optimizer, genome, CMTS) are coordinated by the Adaptive Learning Coordinator (ALC). Individual wing control: 8 independent motors, 8 ODrive S1 controllers, CAN nodes 0–7, one per wing. No paired motor constraints.

Hardware architecture

Parameter	Value
Motors	8 × independent BLDC, one per wing
Controllers	8 × ODrive S1, CAN nodes 0–7
Sensing	Per-wing current at 50 Hz; bus voltage; VSI; IMU
Airframe mass	4.1 kg (MTOW discrepancy — see pre-build items)
Battery	6Ah 22.2V nominal ≈ 133 Wh
Wing span	0.772 m per wing × 8 wings
Wing chord	0.12 m
Total wing area	0.741 m²
Nominal freq	1.87 Hz / 96° amplitude
Re (tip)	~121,584 Re (cruise 9 m/s): ~72,000
Strouhal (cruise)	0.238 (whisper climb: 0.303 — at optimum)

Parameter	Value
SPL at 76m	24.1 dB (eVTOL limit: 65 dB — margin: 40.9 dB)

2. The Five Learning Systems

All five operate independently and are coordinated (not replaced) by the ALC. If the ALC fails, all five continue exactly as before.

AI Observer

`ai_observer_tick()`

Watches sensor stream at 50 Hz. 19 analysis functions. Detects anomalies, patterns, and improvement opportunities. Issues proposals to pilot via ALC communication protocol.

Discovery Selector

`discovery_hypothesis_select()`

162 hypotheses across 44 families. ALC priority queue checked first — observer anomalies immediately redirect evaluation. Respects envelope gate (hover-dependent hypotheses skip when HOVER_CAPABLE=False). Focus keywords and lever-zone auto-routing.

ALSS Optimizer

`_ALSS_SEGMENTS / _ALSS_CURRICULUM`

All 162 hypotheses covered across Classes A–E. Class A: first-flight priority (H-PERF-6 runs before all others). Each segment defines the flight condition, duration, and scoring for one hypothesis evaluation.

Genome

`_MW_* axes (15 total)`

15 genome axes: Roll, Pitch, Yaw, Stability, Soaring, Glide, Turn, Water, Cargo, Competition, Performance, Endurance, Transition, SAR, Resilience. Weighted mechanisms within each axis. Updated after every positive trial. Persisted across flights.

CMTS

`_CMTS_CHAMPIONS (11 contexts)`

Cross-Mode Transfer System. When a configuration in one context beats the champion in another, it promotes via ALC. Silent promotions below the beat threshold; pilot confirmation above it. 11 contexts including regen_efficiency added this session.

3. Adaptive Learning Coordinator (ALC)

The ALC sits alongside the five learning systems as a coordinator. It reads from and writes to a shared state (_ALC_SHARED) that all five systems can access. Communication with the pilot follows the sterile cockpit protocol derived from FAA FAR 121.542 and CRM best practice.

Sterile cockpit protocol

Phase	Status	ALC Communication
pre_flight	**OPEN**	Full briefing — alc_preflight_brief
launch_prep	**STERILE**	No ALC communication
takeoff	**STERILE**	Tier 3 safety alerts only
climb	**STERILE**	Tier 3 safety alerts only
cruise	**OPEN***	Brief operational observations (co-pilot style)
hover_patrol	**OPEN***	Brief operational observations
manoeuvre	**STERILE**	Tier 2/3 only
approach	**STERILE**	Tier 3 only
landing	**STERILE**	Tier 3 only
post_flight	**OPEN**	Full debrief — alc_debrief

* Open* = brief operational observations, 45s minimum gap

CRM three-tier escalation

Tier	Equivalent	Behaviour
Tier 1	Concern	Cruise only. Stated once. Not repeated unless worsens.
Tier 2	Advisory / Discomfort	Any phase. Stated once. Logged.
Tier 3	Unsafe	Any phase. Repeated until alc_ack_alert called.

ALC modalities (registered)

mechanical_wear

Tracks hinge friction, winglet stiffness, winding resistance per wing at 50 Hz. Baseline established after 3 flights. Advisories via CRM tiers. Persists: wear_state.json

anomaly_archaeology

Welford online statistics per lever zone / metric at 2 Hz. Detects accidental improvements >1.5σ from zone mean. Cluster promotion: ≥5 observations across ≥2 flights, +3% advantage. Persists: session_log in alc_session.json

regen_recovery

Monitors regenerative energy return in all braking / descent zones. Estimates regen current from back-EMF. SOC derate above 95%. Persists: regen_state.json

4. Hypothesis Registry

162 hypotheses across 44 families. All queued and covered by ALSS. H-PERF-6 has priority 0 and is always first. Hover-dependent hypotheses are gated by _ENVELOPE['HOVER_CAPABLE'] — set on first flight.

Family	N	First hypothesis label
H-ACC	7	Stability switch — unstable cruise to stable tuck
H-AM	4	Shuttlecock brake — max pressure drag at 90° AoA
H-AQ	3	Whisper cruise — low frequency, high amplitude, St
H-BS	3	Snow surface adaptive takeoff — weight-forward to
H-CE	3	High-altitude amplitude scaling — compensate for t
H-CLB	4	Best rate of climb (V_y) — maximum Ps at 9-10 m/s
H-CR	5	GPS-denied hover stability — cascade tuning for mi
H-DA	7	Density-adaptive hover — real-time ρ estimation fr
H-DIH	6	Geometric dihedral sweep — find optimal restoring
H-EN	3	Sprint to hover — minimum stopping distance from m
H-EV	2	MIL-STD-1797B handling qualities — bandwidth, phas
H-FD	5	Pigeon steep descent — high-dihedral stroke plane
H-FG	3	Conservative flap-glide (3s flap / 2s glide)
H-GS	1	Per-wing gust suppression — individual wing curren
H-HB	5	Hummingbird stroke-plane turn — cascade tilt vecto
H-HL	5	Maximum hover thrust — all 8 wings at peak, clap a
H-HS	4	Drag-minimised maximum sweep — high-speed drag red
H-KW	4	Kestrel body angle equilibrium — stopped wings, pi
H-LB	3	Level-body braking — front/rear cascade differenti
H-MO	3	Maximum sweep at Vne approach — drag minimisation
H-MV	4	Proverse yaw — inside-wing amplitude boost in turn
H-OV	3	80 Jump — cascade angle that maximises simultaneou
H-PDW	3	Stability triangle — proverse / dihedral / weather
H-PERF	7	Top speed trial — maximum level-flight airspeed at
H-PF	6	Fixed-wing pitch feel — aft stick produces proport
H-PH	4	In-phase sprint — collapsed metachronal cascade
H-PREC	5	Trim-optimised cruise — minimum AoA correction at
H-PULL	1	High-load pitch authority — progressive cascade pu
H-RC	4	High-bank turn tightening — aft pitch past 0.7 at
H-REGEN	2	Regenerative descent — wing config that recovers m
H-RES	3	Single wing pair loss — cascade redistribution acr
H-SAR	4	Forward-flight SAR orbit — tight slow circle at V_
H-SE	2	Surface effect mode — WIG + low-altitude flap-glid
H-SK	3	High-AoA power stroke — maximum mid-stroke pitch
H-SOR	6	Glide polar optimisation — minimum sink wing confi
H-SP	2	Nose-down sensor hover — front-pair forward cascad
H-ST	3	Strouhal-optimal cruise frequency (9 m/s)

Family	N	First hypothesis label
H-STOOP	1	Dive dart — speed-adaptive wing area maximises div
H-STORM	7	Gust-damping cascade phase (tandem-wing passive at
H-TR	3	Speed-adaptive pair handoff — progressive thrust/l
H-TX	1	Seamless hover-to-cruise transition — cascade blen
H-VIFF	3	Differential VIFF — outside-wing AoA increase for
H-WC	2	Dorsal clap-and-fling at maximum validated amplitu
H-WD	3	Water departure clap auto-arm — seagull 22% lift s

4.1 New Hypotheses — This Session

ID	Label
H-OV-3	Combined rearward+lateral translation — diagonal, arced lateral,
H-EN-1	Sprint to hover — minimum stopping distance from max speed, casca
H-EN-2	Stoop to lateral translation — convert dive kinetic energy into s
H-EN-3	Direction reversal — rearward flight to forward sprint without st
H-RES-1	Single wing pair loss — cascade redistribution across 7/8 pairs m
H-RES-2	Two wing pair loss — controlled flight maintained with 6/8 pairs,
H-RES-3	Fault detection and adaptation latency — time from wing pair fail
H-AQ-3	Whisper pigeon — reduced-frequency dihedral descent: quieter than
H-FD-5	Pigeon descent with lateral repositioning — descend flat-body whi
H-HL-5	Heavy lift lateral repositioning — translate sideways with suspen
H-REGEN-1	Regenerative descent — wing config that recovers maximum energy i
H-REGEN-2	Regenerative braking — active brake and emergency stop zones retu
H-PERF-6	First-flight performance characterisation — map complete flight e
H-PERF-7	Minimum airspeed refinement — find wing configuration that achiev
H-PREC-2	Heading-stable cruise — cascade phase for zero yaw torque
H-SAR-FF-1	Forward-flight SAR orbit — tight slow circle at V_min for sensor
H-SAR-FF-2	Minimum-speed reversal turn — tightest altitude-stable 180° direc
H-SAR-FF-3	Sensor orbit approach — curved descending approach that keeps the
H-SAR-FF-4	Forward-flight payload drop — minimum-speed pass over target with

5. Genome — 15 Axes

The genome tracks which kinematic mechanisms are currently most effective across 15 independent axes. Weights are updated after every positive trial and normalised to sum to 1.0 per axis. Persisted via persist_save_all.

Axis	Top mechanism	Mechanisms (sample)
ROLL	roll_amp	roll_amp, roll_aoa, roll_dih, roll_phase
PITCH	pitch_freq	pitch_aoa, pitch_dih, pitch_freq, pitch_phase
YAW	yaw_phase	yaw_amp, yaw_dih, yaw_h26, yaw_phase
STABILITY	geom_dihedral	geom_dihedral, proverse_yaw, speed_adapt_dih, stroke_dihedral
SOARING	glide_aoa	glide_aoa, glide_sweep, maccready_speed, slow_flap_assist
GLIDE	min_sink_aoa	best_glide_aoa, drag_minimise, min_sink_aoa, span_extension
TURN	stroke_plane_tilt	inertial_damping, inner_outer_asymm, lev_drag_precision, per_stroke_vector
WATER	clap_wig_synergy	clap_wig_synergy, front_rear_phase, stroke_Q_asymm
CARGO	max_thrust_fm	ballistic_climb, max_thrust_fm, payload_trim, pendulum_damp
COMPETITION	gps_denied_drift	acoustic_signature, alt_loiter, cargo_range, gps_denied_drift
PERFORMANCE	top_speed	cruise_efficiency, hover_efficiency, intent_latency, precision_hold
ENDURANCE ★	sustained_wh_per_km	correction_frequency, current_variance, sustained_wh_per_km, thermal_budget
TRANSITION ★	blend_smoothness	altitude_hold, blend_smoothness, pitch_continuity, speed_accuracy
SAR ★	area_coverage_rate	acoustic_minimum, area_coverage_rate, payload_delivery, precision_hover
RESILIENCE ★	single_pair_loss	fault_detect_speed, pair_pair_loss, proverse_redundancy, single_pair_loss

★ New axis added this session

6. Regenerative Energy Recovery

Philosophy: any energy that would become heat in motor windings during braking or descent returns to the battery instead. The primary goal is always flight quality. Regen is what happens to the energy that would otherwise be wasted.

Context	Lever zone	Mechanism	Est. power
Active brake	6–16%	Cascade reversal back-EMF	Highest I_regen
Descent	16–28%	Dihedral drag + gravity	~5–8 W sustained
Glide/soar	28–38%	Airspeed-driven back-EMF	~2–5 W sustained
Emergency stop	0–6%	Max cascade reversal	Brief high pulse

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ODrive configuration required: odrv0.config.dc_bus_max_negative_current = -5.0 | Battery SOC derate active above 95% SOC | Efficiency estimate: 28% (realistic for reciprocating geometry)
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7. Flight Envelope Characterisation

MTOW_KG = 3.448 kg vs M_AIRFRAME_KG = 4.1 kg: the airframe mass exceeds the MTOW estimate. Hover capability is uncertain until measured. H-PERF-6 runs as the first hypothesis on first flight and resolves this with real data. All subsequent hypothesis selection is conditioned on the measured result.

Envelope gate logic

HOVER_CAPABLE = None	Both hover and FF hypotheses run (measuring)
HOVER_CAPABLE = True	All 162 hypotheses active
HOVER_CAPABLE = False	Hover-dependent hypotheses gated off; H-SAR-FF-1/2/3/4 primary

Forward-flight SAR mode (active regardless of hover)

ID	Description
H-PERF-6	First-flight envelope characterisation — priority 0, runs before all others
H-PERF-7	V_min refinement — find wing config achieving lowest sustainable airspeed
H-PREC-2	Short-field landing — minimum touchdown velocity in flare
H-SAR-FF-1	Slow orbit — tight banked circle at V_min for sensor dwell
H-SAR-FF-2	Min-speed reversal turn — tightest 180° turn above V_min (BK only)
H-SAR-FF-3	Sensor orbit approach — descending orbit keeping target in sensor FOV
H-SAR-FF-4	Forward-flight payload drop — per-wing amplitude reduction at release

8. Ground Station Commands — New This Session

Command	Purpose
`alc_preflight_brief`	Pre-flight learning briefing — what was learned, what is queued
`alc_debrief`	Post-flight debrief — full session narrative including regen, wear, archaeology
`alc_status`	Current ALC phase, alerts, queue, modalities
`alc_set_flight_phase`	Set sterile cockpit phase: pre_flight, cruise, approach, etc.
`alc_ack_alert`	Acknowledge pending tier-3 safety alert
`learning_rate good/neutral/poor`	Pilot satisfaction signal — reinforces or reduces genome weight
`wear_status`	Per-wing hinge friction, winglet stiffness, winding resistance, stroke counts
`archaeology_status`	Accidental improvement clusters — pending and promoted
`regen_status`	Session and all-time regenerative energy totals by zone
`regen_config_advice`	ODrive configuration steps to enable regeneration
`resilience_status`	Wing pair fault status and resilience genome state
`thermal_advisory_status`	Current thermal detection and H-SOR-6 advisory state
`autopilot_deferred_status`	List of autopilot-phase items with development order
`combined_translation_status`	H-OV-3 combined translation mode and state
`envelope_status`	Flight envelope — HOVER_CAPABLE, V_min, orbit radius
`set_hover_capable true/false`	Manually set hover capability from pilot observation

9. Autopilot Deferred Items

The distinction is architectural: background kinematics decide how the wings move to express pilot intent. Autopilot decides where the aircraft goes. All items below require path planning or trajectory management.

Item	Rationale for deferral
H-SAR-1	Expanding square search pattern — path planning required
H-SAR-2	Terrain contour search — path planning required
H-SAR-3	Confined space approach — combined autopilot + sensor fusion
H-SAR-4	Victim vector approach — path planning from sensor fix
H-SOR-7	Thermal exit sprint — thermal exploitation autopilot decision
RTL	Return to launch — autopilot post first flight
POSITION_HOLD	GPS/vision position hold — autopilot post first flight
COURSE_HOLD	Course hold with wind correction — autopilot
PATTERN_FLIGHT	Autonomous search patterns — autopilot
ORBIT_AUTONOMOUS	Autonomous SAR orbit — software chooses orbit centre, radius, and entry/exit; pilot-commanded orbit kinematics are in H-SAR-FF-1 (active now). Autopilot phase.
LAWNMOWER_PATTERN	Autonomous lawnmower search — software chooses track spacing, flies parallel tracks, reverses at track ends. Turn kinematics are in H-SAR-FF-2 (active now). Autopilot phase.
DESCENDING_APPROACH_AUTO	Autonomous descending approach to target — software guides aircraft along computed glidepath to victim location. Pilot-flown approach kinematics are in H-SAR-FF-3 (active now). Combines with H-SAR-4 in autopilot phase.

10. Post-First-Flight Development Queue

Directional microphone array

SAR victim detection from altitude. Humanitarian application of the Blue Thunder concept. Pointed downward to detect and triangulate survivor voice or whistle. No hypothesis written yet.

Weather / low-visibility family

Rain, crosswind gusts with precipitation, wet wing surface aerodynamics, optical sensor degradation. Needs baseline normal-conditions data first.

Turbulence regenerative charging

Extend H-GS-1 gust suppression to harvest overdrive condition energy. Joint objective: suppression quality + energy recovery. Requires H-REGEN-1/2 validated first.

Environmental characterisation

Per-site wind pattern learning. Deferred to autopilot phase — may connect to navigation and national weather data.

Pilot behaviour modelling

Deferred to autopilot phase.

Voice communication (AI client)

Earpiece + STT/TTS for post-flight debrief conversation. Deferred to post-first-flight / autopilot phase.

11. Pre-Build Physics Audit — Open Concerns

From the adversarial physics audit conducted this session. The design is grounded in established science. These are engineering concerns, not conceptual failures.

HIGH **MTOW vs airframe mass**

MTOW_KG=3.448 but M_AIRFRAME_KG=4.1. Airframe alone exceeds MTOW. Hover performance marginal. Resolve before build: confirm build target mass or revise wing geometry / frequency ceiling.

MED-HIGH **Structural load limit unverified**

STOOP_MAX_WING_LOAD_G=4.0 is an estimate (TC-05 will calibrate). Pre-flight static load testing of TPU winglet hinges across temperature range required before any stoop or high-speed manoeuvres.

MEDIUM **Tandem interference not in performance model**

Literature shows 7–20% lift reduction from tandem interference. Per-wing control actively manages this (better than literature suggests), but baseline estimates should be discounted ~10% until flight data confirms.

MEDIUM **Winding resistance formula unit check**

Wear modality back-EMF calculation may have RPM/rad unit inconsistency. Verify against ODrive telemetry before trusting baseline wear data.

LOW-MED **Laminar separation bubble — no hypothesis**

At Re 70,000–120,000, LSB formation at stroke reversal is real. TPU compliance is probably protective. Genome will respond empirically but without naming the mechanism. Add LSB hypothesis after first flight.